

Contact

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marcialcariqueo (LinkedIn)

Top Skills

Population Pharmacokinetics

PK/PD Modeling

Real World Evidence (RWE)

Languages

Español (Native or Bilingual)

Catalán (Elementary)

Inglés (Professional Working)

Certifications

Primeros Pasos en la Docencia USS
Online: Una Guía para los Docentes

Tools for Data Science V2

Python for Data Science and AI

Data Science Orientation

Honors-Awards

Premio al mejor trabajo de
investigación modalidad de
presentación póster.

Premio al mejor trabajo de
investigación modalidad de
presentación oral

Publications

Recomendaciones SOCHIMI para el
manejo inicial de la sepsis

Síndrome de Dress. Eritema
cutáneo, fiebre y hepatitis asociado
a lamotrigina. Reporte de dos casos
y revisión de la literatura

Neurocritical Care Society News

Analgesia, Sedation, Neuromuscular
Blockade and Delirium Management
Practices in Chilean Intensive Care
Units]

Marcial Cariqueo Arriagada

Clinical Pharmacist | PhD Candidate, Translational Medicine (UB
· IDIBAPS) | PK/PD & Pharmacometrics in Severe Respiratory
Infections | Critical Care · Clinical Trials

Barcelona, Catalonia, Spain

Summary

Clinical pharmacist and PhD candidate in Translational Medicine (Universitat de Barcelona), specialized in pharmacometrics: population PK/PD modeling of anti-infectives in severe respiratory infections. 16 years across hospital pharmacy, critical care and clinical research — 7 of them delivering therapeutic drug monitoring (TDM) and PK/PD-guided dosing in the ICU. Now at IDIBAPS – Hospital Clínic de Barcelona (Prof. Antoni Torres's group), working from preclinical PK/PD models to real-world evidence from large pneumonia cohorts, and coordinating multicenter clinical trials under ICH-GCP. Core expertise: Pharmacometrics · Population PK (NONMEM, PsN) · PK/PD · Model-Informed Drug Development (MIDD) · Clinical Trials (ICH-GCP) · Real-World Data · Anti-infectives · TDM · R · Python · REDCap Publications in the Journal of Antimicrobial Chemotherapy and European Respiratory Review, among others.

Experience

Institut d'Investigacions Biomèdiques August Pi i Sunyer (IDIBAPS)
1 year 11 months

Predoctoral Research Fellow | Translational Medicine
March 2026 - Present (6 months)
Barcelona

Clinical Study Coordinator & Sub-Investigator
October 2024 - Present (1 year 11 months)
Barcelona

Coordinating clinical research studies in respiratory infections, pneumonia, and vaccine effectiveness at Hospital Clínic Barcelona. Responsible for site-level study operations, patient recruitment, protocol implementation, and clinical data management.

- RECAP20 — Pneumococcal pneumonia and PCV20
- Id.DRIVE — Respiratory infections and vaccine effectiveness, including COVID-19 and RSV
- Patient screening, recruitment, and protocol implementation
- REDCap-based clinical data management and quality control

Centro de Investigación Biomédica en Red CIBER

Research Collaborator – CIBERES (CB06/06/0028)

September 2025 - Present (1 year)

Spain

Research Collaborator within the CIBER de Enfermedades Respiratorias (CIBERES), Research Group CB06/06/0028, contributing to translational and clinical research in respiratory infections, pneumonia, and critical care. Collaborative activities include clinical research, study coordination, data analysis, and translational research projects within the Hospital Clínic Barcelona–IDIBAPS research environment.

Hospital Evangèlic de Barcelona

Hospital Pharmacy Assistant

August 2022 - January 2024 (1 year 6 months)

Barcelona

Supported hospital pharmacy operations at Nou Hospital Evangèlic, contributing to the safe and efficient management of medications within a patient-centered hospital environment. Involved in pharmaceutical logistics, inventory management, medication safety, and quality processes supporting daily clinical care.

- Medication inventory, stock control, and pharmaceutical logistics
- Medication safety and support of hospital dispensing workflows
- Compliance with established quality and ISO procedures
- Collaboration within a multidisciplinary healthcare environment

Universidad de Chile

7 years 8 months

Academic Instructor

January 2015 - August 2022 (7 years 8 months)

Chile

Delivered undergraduate and postgraduate teaching in clinical pharmacy and pharmacotherapy, integrating evidence-based medicine with real-world clinical practice.

- Teaching and mentoring of pharmacy students and healthcare professionals
- Clinical pharmacotherapy and evidence-based medicine education
- Development and delivery of academic and clinical training activities

Critical Care Program Coordinator

January 2022 - June 2022 (6 months)

Chile

Coordinated the Critical Care training program, overseeing academic activities and supporting multidisciplinary education in the management of critically ill patients.

Hospital Clínico Universidad de Chile

7 years

Critical Care Pharmacist

August 2015 - July 2022 (7 years)

Santiago

Provided specialized clinical pharmacy services for critically ill patients, integrating clinical pharmacokinetics, therapeutic drug monitoring (TDM), and individualized pharmacotherapy optimization within a multidisciplinary ICU team.

- TDM and PK-guided dose optimization for complex and critically ill patients
- Antimicrobial therapy optimization using PK/PD principles and microbiological data
- Individualized dose adjustment according to renal function and evolving clinical status
- Prospective pharmacotherapy review during multidisciplinary ICU rounds
- Medication safety, pharmacovigilance, and prevention of drug-related problems
- Contribution to antimicrobial stewardship and pharmacy quality-improvement initiatives

Clinical Research Coordinator

January 2021 - June 2022 (1 year 6 months)

Chile

Coordinated industry-sponsored and academic clinical studies at Hospital Clínico Universidad de Chile, supporting study implementation, patient recruitment, clinical data management, and site-level research activities.

- Coordination of patient screening, recruitment, and study procedures

- Source data review and eCRF management in compliance with study protocols
- Safety reporting support and communication with sponsors and clinical teams
- Coordination of a multicenter study evaluating biomarkers for early detection of acute kidney injury

Hospital Clínic de Barcelona

1 year 1 month

Population Pharmacokinetics Fellowship

January 2019 - January 2020 (1 year 1 month)

Greater Barcelona Metropolitan Area

- Developed a pharmacokinetic model of tigecycline in critical care patients with Liver disease
- Conducted research on Population Pharmacokinetics at Hospital Clínic de Barcelona.

Professional Training

January 2019 - December 2019 (1 year)

Barcelona y alrededores, España

- Trained in Hospital Pharmacy, Clinical Pharmacokinetics, and Clinical Pharmacy in Critical Patients.

3M Health Care

4 years

Intravenous Therapy Consultant

2014 - 2018 (4 years)

Chile

- Provide advice and scientific support in the area of medication administration.
- Continuous education for healthcare professionals.

Key Opinion Leader

2014 - 2018 (4 years)

Chile

- KOL in Intravenous Therapy

AstraZeneca

1 year

Analyst of Pharmacovigilance & Regulatory Affairs

2014 - 2015 (1 year)

- Managed pharmacovigilance activities including PSUR and DSUR reports for patient safety at AstraZeneca in Chile.

- Developed strategies for medical information dissemination and regulatory affairs in clinical trials.
- Collaborated with cross-functional teams to ensure compliance and safety standards were met.

Medical Information Specialist

2014 - 2015 (1 year)

Chile

Clínica Alemana de Santiago

4 years 7 months

Critical Care Pharmacist

November 2011 - November 2014 (3 years 1 month)

Región Metropolitana de Santiago, Chile

Managed pharmacotherapeutic monitoring for ICU patients, developed protocols in Clinical Pharmacy, and provided continuous training for healthcare professionals.

- Standardized pharmacotherapeutic processes to ensure quality patient care.
- Developed and implemented protocols to enhance medication management.
- Provided ongoing training to physicians and nurses to improve medication safety and efficacy.

Drug Safety Specialist

May 2010 - November 2014 (4 years 7 months)

Región Metropolitana de Santiago, Chile

Led the development of the institutional pharmacovigilance program at Clínica Alemana de Santiago, resulting in improved drug safety practices.

- Conducted training sessions for institution personnel and organized seminars to enhance awareness in drug safety.
- Established recording systems and implemented quality indicators to ensure compliance with regulatory standards.

IV Pharmacist Inpatient

May 2010 - October 2011 (1 year 6 months)

Región Metropolitana de Santiago, Chile

- Planned, validated, and supervised the preparation of total parenteral nutrition and intravenous drugs at Clínica Alemana de Santiago.
- Ensured quality assurance and supply control in the manufacturing of cytostatic drugs.

- Inspected processes and finished products to maintain high standards of pharmaceutical care.

Education

Universitat de Barcelona

Doctor of Philosophy, Medicine and Translational Research · (March 2026)

IT Academy de Barcelona Activa

Data Analytics · (January 2024 - July 2024)

Universidad de Chile

Master's degree, Pharmacology · (March 2018 - January 2022)

Universidad de Chile

Diploma en Bioestadística y Epidemiología Aplicada · (September 2016 - January 2017)

Pontificia Universidad Católica de Chile

Doctor of Pharmacy - PharmD, Pharmacy · (March 2004 - December 2009)